

MTH301-Calculus-II

Solved MCQS for Midterms papers

Solved by JUNAID MALIK and Team



AL-Junaid Institute

Contact no: 03041659294



LMS Handling

Subject Enrollment

- 1) Online classes available
HTML, CSS, Javascript
Phyton..etc
- 2) Solved Graded Activities
 - Assignment's
 - Quizzes
 - GDB's
- 3) Solved Quiz File's
- 4) Short Note's
- 5) Past paper's &
Current paper's

Website Link

vulmshelp.com

Final project CS619

- 1) SRS
(Software Requirement's
Specification)
- 2) DD
(Design Document)
- 3) Test phase + viva
- 4) Viva preparation
- 5) Final Deliverable

Gmail

junaidfazal08@gmail.com

AL-JUNAID INSTITUTE GROUP

Every point in three dimensional space can be described by ----- coordinates

- **Three**

What are the direction cosines for the line joining the points (1, 3, 2) and (7, -2, 3)?

- $\frac{6}{\sqrt{62}}, \frac{-5}{\sqrt{62}} \text{ and } \frac{1}{\sqrt{62}}$

The angles which a line makes with positive x ,y and z-axis are known as -----

Direction angles

- **Plane parallel to xz-plane**

Domain of the function $f(x, y, z) = \sqrt{x^2, y^2, z^2}$ is

- **Entire 3D-Space**

Suppose $f(x, y) = xy - 2y$ where $x^2 = +3t$ and $y = 2t$. Which one of the following is true?

- $\frac{df}{dt} = -16t - t$

For a function $f(x, y, z)$, the equation $\partial^2 \partial^2 \partial^2 x f + \partial^2 \partial^2 \partial^2 y f + \partial^2 \partial^2 \partial^2 z f = 0$ is known as -----
--- --

- **Laplace's Equation (Ali)**

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

- **4.2**

AL-JUNAID INSTITUTE GROUP

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- **Radius of inscribed circle in R**

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are -----

- **Perpendicular**

Two surfaces are said to intersect orthogonally if their normals at every point common to them are -----

- **In opposite direction**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- **Radius of inscribed circle in R**

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are -----

- **Perpendicular**

Two surfaces are said to intersect orthogonally if their normals at every point common to them are -----

- **In opposite direction**

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ and $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ If $D > 0$ and $f_{xx}(x_0, y_0)$

- **Relative maximum at**

AL-JUNAID INSTITUTE GROUP

Let the function $f(x, y)$ has continuous second-order partial derivatives (f_{xx} , f_{yy} and f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$. If $D = 0$ then -----

- **No conclusion can be drawn.**

Every real number corresponds to _____ on the co-ordinate line.

- **A unique point**

There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- **Set of real numbers**

Which of the following is associated to each point of three dimensional spaces?

- **A natural Number**

All axes are positive in _____ octant.

- **First**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- **Radius of inscribed circle in R**

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are _____.

- **Perpendicular**

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R , then $f(x, y)$ has both _____ on R .

- **Absolute maximum and absolute minimum value**

Let the function $f(x, y)$ has continuous second-order partial derivatives (f_{xx} , f_{yy} and f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has _____ .
Relative maximum at

AL-JUNAID INSTITUTE GROUP

- **Relative maximum at**

Let the function $f(x, y)$ has continuous second-order partial derivatives f_{xx} , f_{yy} and f_{xy} in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if

if $D = 0$ then _____.

- **No conclusion can be drawn.**

Every real number corresponds to _____ on the co-ordinate line

- **Infinite number of points**

There is one-to-one correspondence between the set of the points on co-ordinate and _____

- **Set of natural number**

Which of the following is associated to each point of three dimensional spaces?

- **A natural number**

All axes are positive in _____ octant

- **First**

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$ when are its cylindrical co-ordinates?

- **$\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$**

Suppose $f(x, y) = xy - 2y^2$ Where $x = 3t + 1$ and $y = 2t$ which one of the following is true ?

AL-JUNAID INSTITUTE GROUP

- $\frac{df}{dt} = -4t + 2$

Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = t(r, s)$ then by chain rule

$$\frac{\partial w}{\partial r} =$$

- $\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \frac{\partial z}{\partial s}$

Is the function $f(x, y)$ continuous at origin ? if not ,why?

$$f(x, y) = \begin{cases} 0 & \text{if } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{otherwise} \end{cases}$$

- F(x, y) is not defined**

Is the function $f(x, y)$ continuous at origin ? if not ,why?

$$f(x, y) = \begin{cases} 0 & \text{if } (x, y) \neq 0 \\ 1 & \text{if } (x, y) = 0 \end{cases}$$

- F(x, y) is not defined**

Let R be a closed region in two dimensional spaces. What does the double integral over R calculate?

- Radius of inscribed circle in r**

Every real number corresponds to _____ on the co-ordinate line.

- Infinite number of points**

There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of natural numbers**

AL-JUNAID INSTITUTE GROUP

Which of the following is associated to each point of three dimensional spaces?

- **A natural Number**

All axes are positive in _____ octant

- **First**

By Extreme Value Theorem, if a function $f(x, y, z)$ is continuous on a closed and bounded set R , then $f(x, y, z)$ has both _____ on R .

- **Absolute maximum and absolute minimum value**

Let x be any point on co-ordinate line. What does the inequality $-3 < x < 1$ means?

- **The set of all real numbers between -3 and 1**

Which of the following number is associated to each point on a co-ordinate line?

- **A real number**

Which of the following set is the union of set of all rational and irrational numbers?

- **Set of real numbers**

----- planes intersect at right angle to form three dimensional space.

- **Three**

each point of domain, the function -----

- **Is defined**

Suppose $f(x, y) = xe^{3xy}$ Which one of the following is correct?

AL-JUNAID INSTITUTE GROUP

- $\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

Suppose $f(x, y) = xe^{3xy}$ Which one of the following is correct?

- $\frac{\partial f}{\partial y} = x^4 e^{xy}$

Suppose $f(x, y) = 2xy$ where $x = t^2 + 1$ and $y = 3 - t$ which one of the following is true?

- $\frac{\partial f}{\partial t} = -6t^2 + 12t - 2$

i, j and k are unit vectors in the direction of x -axis, y -axis and z -axis respectively. Suppose that $\vec{a} = 2i + 5j - k$

- $\sqrt{30}$

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

- $1, -4$ and -1

Which of the following is geometrical representation of the equation $y = x^2$, in three dimensional space?

- **Parabola**

Which of the following is the interval notation of real line?

- $(-\infty, +\infty)$

What is the general equation of parabola whose axis of symmetry is parallel to y -axis?

- $y = ax^2 + bx + c$ ($a \neq 0$)

AL-JUNAID INSTITUTE GROUP

Which of the following is geometrical representation of the equation $y=4$, in three dimensional space?

- A point on y-axis

Plane is an example of -----

- Surface

Function $f(x, y) = \sqrt{y-x}$ is continuous in the region ----- and discontinuous elsewhere.

- $X \leq y$

At critical points function can assume

- Zero value

A plane can be perfectly determined by

- One point and normal vector

Intersection of two straight lines is -----

- Point

Plane is a ----- surface.

- Two-dimensional

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t \hat{i} + (1 - \cos 2t) \hat{j}.$$

- $X = \sin^2 t, y = 1 - \cos 2t, z = 0$

AL-JUNAID INSTITUTE GROUP

What are the parametric equations that correspond to the following vector equation?

$$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\hat{j} + \sin 3t\hat{k}$$

- $X=2t-1, y=-3\sqrt{t}, Z=\sin 3t$

What is the derivative of following vector-valued function?

$$\vec{r}(t) = (\cos 5t, \tan t, 6 \sin t)$$

- $\vec{r}(t) = (-5 \sin 5t, \sec^2 t, 6 \cos t)$

The following differential is exact $dz = (x^2 y + y)dx - xdy$

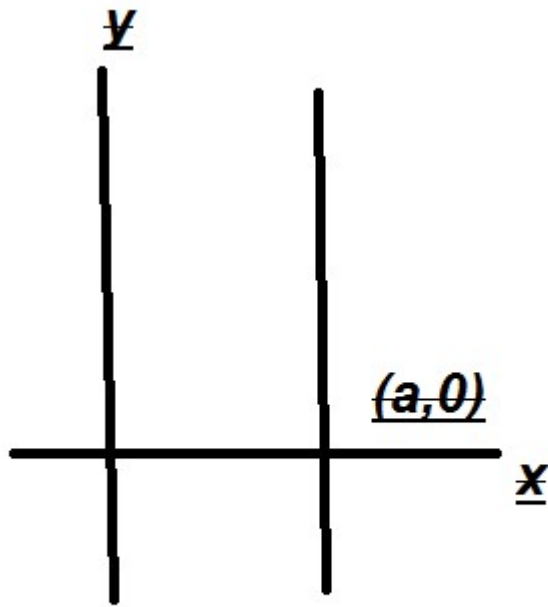
- False

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

- $\int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots \frac{5}{6} \frac{3}{4} \frac{1}{2}$

Match the following equation in polar co-ordinates with its graph. r a where a is an arbitrary constant

AL-JUNAID INSTITUTE GROUP



If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

- Y-axis

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

- Pole

What is the amplitude of a periodic function defined by $f(x) = \sin 3x$

- 0

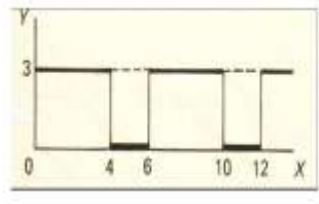
What is the period of a periodic function defined by $f(x) = 4\cos 3x$?

- $\frac{2\pi}{3}$

AL-JUNAID INSTITUTE GROUP

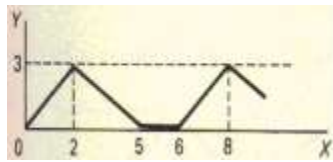
Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$



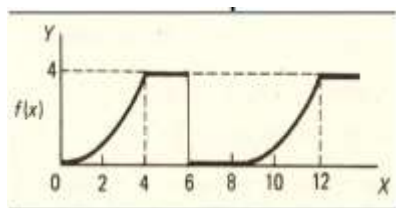
•

What is the period of periodic function whose graph is as below?



6

What is the period of periodic function whose graph is as below?



• 8

Let L denotes the Laplace Transform. If $L\{F(t)\} = f(s)$ where s is a constant. And

$\lim_{x \rightarrow \infty} \left(\frac{f(t)}{t} \right)$ Exists then which of the following equation holds?

• $L\left(\frac{f(t)}{t}\right) = \int_s^{\infty} f(s) ds$

Which of the following is Laplace inverse transform of the function $f(s)$ defined by

$$f(s) = \frac{3}{s-2} - \frac{2}{s} ?$$

• $3e^{2t} - 2$

AL-JUNAID INSTITUTE GROUP

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

- Midpoint of the line joining these two points

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy -plane. If $(P dx + Q dy)$ is an exact

differential then $\int_C (P dx + Q dy) =$

- Zero

What is Laplace transform of the function $F(t)$ if $F(t) = t$?

- $L\{t\} = \frac{1}{s}$

What is the value of $L\{e^{5x}\}$ if L denotes Laplace transform?

- $L\{e^{5x}\} = \frac{1}{s-5}$

Evaluate the line integral $\int_C (3x + 2y)dx + (2x - y)dy$ the line segment from $(0, 0)$ to $(0, 2)$.

- 2

Evaluate the line integral $\int_C (2x + y)dx + (x^2 - y)dy$ where C is the line segment from $(0, 0)$ to $(2, 0)$.

- 0

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

2, -3 and 20

AL-JUNAID INSTITUTE GROUP

if $R = \{(X, Y) / 0 \leq X \leq 2 \text{ and } 1 \leq Y \leq 4\}$, then $\iint_R (6x^2 + 4xy^3) dx =$

- $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx dy$

----- Planes intersect at right angle to form three dimensional spaces.

- Three

If the positive direction of x, y axes are known then ----- the positive direction of z-axis.

- Left hand rule tells

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- $\frac{7}{\sqrt{2}}$

The equation $ax + by + cz + d = 0$, where a, b, c, d are real numbers, is the general equation of which of the following?

- Plane

Domain of the function $f(x, y) = \sqrt{y - x^2}$ is

- $y \geq x^2$

What is the period of a periodic function defined by $f(x) = \sin x/2$?

- 4π

The graph of an odd function is symmetrical about -----

AL-JUNAID INSTITUTE GROUP

- **Origin**

The path of integration of a line integral must be -----

- **Continuous and single-valued**

Sign of line integral is reversed when -----

- **Direction of path of integration is reversed.**

What is Laplace transform of the function $f(t)$ if $f(t) = \sin 3t$

- $L\{\sin 3t\} = \frac{3}{s^2 + 9}$

π is an example of -----

- **Irrational numbers**

Straight line is a special kind of -----

- **Curve**

An ordered triple corresponds to ----- in three dimensional spaces.

- **A unique point**

The angles which a line makes with positive x, y and z-axis are known as -----

- **Direction angles**

Is the function $f(x, y)$ continuous at origin? If not, why? $f(x, y) = 4xy + \sin 3x^2y$

- **$f(x, y)$ is continuous at origin**

Two vectors are opposite vectors if they have -----.

- **both (a) and (c)**

AL-JUNAID INSTITUTE GROUP

If two dogs are pulling a bone with force=20 Newtons in opposite direction, the resultant force is _____

- Zero Newton

If the velocity (V1) of a car is 20m/s and velocity (V2) of bike is 40m/s, where both velocities are southward then which of the following would be the dot product of these velocities?

- -800

Limit of a function may exist at the point where it not defined

- False

Limit of a function may exist at the point where it not defined

- False

A _____ of continuous functions is continuous

- All (a), (b) and (c)

A quotient of continuous function is continuous, except where the denominator is _____

- Zero

Intersection of two surfaces is a _____ in three dimensional space Curve

- Face the parabola ($y=ax^2 + bx + c$) is opening downward when $A<0$

Which coordinate system uses two distances and one angle?

- Cylindrical

AL-JUNAID INSTITUTE GROUP

Which of the following is a two dimensional coordinate system

- Polar coordinate system

Limit of a function exist may exist at the point where it not defined

- False

The number square root of 2 is

- None of these /1.41

A plane $z=3$ is _____ to XY plane.

- Parallel

Value of the function $f(x,y,z)=xyz$ at point $(2,2,2)$ is

- 8

The number $3/11$ is

In rectangular coordinate system the point $(-3,2)$ is in

- Second quadrant

Which of the following function is not the function of one variable?

- $m=g(s) + \text{content}$

XZ- Plane consists of all points

- where $X=0$

Value of the function $f(x,y,z)=x^2 + y^2 + z^2$ at the point $(0,1,0)$ is

- 1

If $w=f(x,y,z)$ where $x=g(t)$, $z=h(T)$ then

- $Dw/dt = @wz/@xdt + @wdy/@ydt + @wdz/@zdt$

If f and its partial derivatives of the first, second and third

orders are continuous on an open set, then at each point of the

set _____

- $@^2f/@y^2@x = @^2f/@y@x@y = @^2f/@x@y^2$

AL-JUNAID INSTITUTE GROUP

All axis are negative in ----- octant

- **Eight**

In three dimensional space, intersection of two surfaces is a -----

- **Plane**

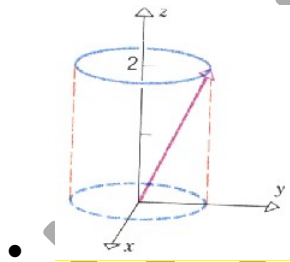
$$\frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y} + \frac{\partial^2 f}{\partial^2 z} = 0$$

For a function $f(x, y, z)$, the equation is known as -----

- **Laplace's Equation**

Match the following vector-valued function with its graph.

$$r(t) = \cos t \hat{i} + \sin t \hat{j} + 2\hat{k} \quad \text{and} \quad 0 \leq t \leq 2\pi$$



What is the derivative of following vector-valued function?

- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

What is the derivative of following vector-valued function?

$$\vec{r}(t) = (e^{t^2}, t^2, \sec 2t)$$

AL-JUNAID INSTITUTE GROUP

- $\vec{r}'(t) = (2te^{t^2}, 2t, 2 \sec 2t \tan 2t)$

The following differential is exact $dz = (x^2 + y^2) dx - 2xy dy$

- **False**

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

- $$\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \dots \frac{5}{6} \frac{3}{4} \frac{1}{2}$$

If $a > 0$, then the equation, in polar co-ordinates, of the form $r^2 = a^2 \cos 2\theta$ represent which of the following family of curves?

- **Lemiscate**

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

- $\vec{F} = 0$

If $\lim_{(x,y,z) \rightarrow (x_0,y_0,z_0)} f(x,y,z) = L_1$ then

- $\lim_{(x,y,z) \rightarrow (x_0,y_0,z_0)} cf(x,y,z) = cL_1$ If c is constant

if $\lim_{(x,y) \rightarrow (x_0,y_0)} F(x,y) = L_1$ $\lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = L_2$

- $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y)/g(x,y) = L_1/L_2$; L_2 not equal to 0

AL-JUNAID INSTITUTE GROUP

The partial derivative with respect to x of $f(x,y) = 3x^2y + 2y^2$ is _____

- $6xy + 2y^2$

Spherical coordinate ϕ is related to the cylindrical coordinate as _____

- $\phi = \theta$

How many real numbers exist between 1 and 5

- Infinity

Value of the function $f(x,y,z) = \text{square root } x^2 + y^2 + z^2$ at the point (0,3,0) is _____

- 3

The number pi is.

- Irrational

Length of magnitude of a unit vector is _____

- 1

The product of continuous

- functions is continuous True

$F(x,y,z) = \text{square root } x^2 + y^2 + z^2$

- $[0, +\infty)$

The product of continuous

- function is continuous True

Range of the function $y = 3x^2 - 2$ is

AL-JUNAID INSTITUTE GROUP

- All real numbers with $y \geq -2$

if the partial derivative (w.r.t 'x' or 'y') at any point on the surface " $z=f(x,y)$ " is zero, then tangent plane at that point will be _____ to XY- plane

- parallel

Two vectors A and B are _____ if and only if their scalar

- is equal to zero. Perpendicular

The partial derivative with respect to x of $f(x,y) = 3x^2y + 2y^2$ is _____

- $6xy$

The chain rule can be taken as the derivative of the order function and multiplying it items the derivative of the _____

- The inner function

Two vectors are opposite vectors if they have _____

- Both (A) and (c)

Magnitude of the force vector $F = -30$ Newton, is _____

- ewton

Two vectors A and B are _____ if and only if their scalar product is equal to zero

- Perpendicular

Which of the following should be the scalar multiple (A) such that force ; $F_1=20\text{N}$ and $F_2 = -60\text{N}$ satisfy the

- $A=-1/3$

A zero vector has magnitude equal to zero and _____

- Direction Arbitrary

AL-JUNAID INSTITUTE GROUP

The chain rule states that the derivative of $f(g(x))$ is _____

- $f(g(x)) \cdot g'(x)$

The order of differentiation in an n th order partial derivative can be change without affecting the final result whenever the function and all its partial derivatives of order less than or equal to ' n ' are _____

- Continuous

For which of the following values of ' y ' the partial derivative of $z = 2/y(x-1)$ EXIST

- $\mathbb{R} - \{1, 0\}$

To find the limit of $f(x,y) = x^2 + y/x^2$ at $(0,0)$ if we approach $(0,0)$ through the line x -axis, we get the value

- 1

Value of money is an example of _____

- Quantity Scalar

IF $y = \sin t$ and $t = e^{xu}$ then which of the following is partial derivative of w.r.t. x

- $\cos t$

The function $z = F(x,y)$ represents a _____ in

- space Line

AL-JUNAID INSTITUTE GROUP

If two dogs are pulling with force=20Newtons in opposite direction, then the resultant force is _____

- **Zero Newton**

Limit of a function may exist at the point where it not

- **defined False**

A _____ of continuous function is continuous

- **All options**

The function e^{xy} is continous

In whole region of xy plane

A quotient of continuous function is continuous, expect where the denominator is _____

- **Zero**

if the partial derivative (w,r,t 'x' or 'y') at any point on the surface " $z=f(x,y)$ is zero, than tangent plane at that point will be parallel to _____

- **XY plane**

Two determine the position of any object in a room we need how many coordinates.

- **3(3 dimension)**

AL-JUNAID INSTITUTE GROUP

Which of the following is a two dimensional coordinate system

- Cylindrical coordinate system

Which of the following is the representation of spherical coordinates

- (ρ, θ, ϕ)

In cartesian coordinate system all axes are at _____ degree to each other

- 90

intersection of two surfaces is a _____ in three dimensional space

- Curve

The function $f(x,y,z) = (x^2 + yz)/$

- $(xy+z) \quad Xy+z=0$

For which of the following values of 'y' the partial derivative of $z=2/y(x-1)$

- $\mathbb{R}-\{1,0\}$

YZ- plane of all points where

- $X=0$

AL-JUNAID INSTITUTE GROUP

Domain of the following function $y = x / 3x - 4$

- All real number

Cylindrical coordinate system is used to the position of any

- object is Any dimension

To express the location of any object on earth which coordinate system is used

- Cylindrica coordinate system

Face of the parabola ($y - ax^2 + bx + c$) is opening downward

- when $A < 0$

Cylindrical coordinate “y” is related to the Cartesian coordinate as

- $R = \cos \theta$

Which coordinate system uses two distances and one angle?

- Rectangular

Which of the following function is not the function of one variable $y =$

- $f\{x, y\}$

Which of the following is a two dimensional coordinate system

- Cylindrical coordinate system

The chain rule can be taken as the derivative of the outer function and multiplying it times the derivative of the _____

- The inner function

AL-JUNAID INSTITUTE GROUP

The rule is called the chain rule because we use it to take derivatives of

_____ by changing together their derivative

- **Composite function**

The partial derivative at any point on the surface represents the slope of tangent _____ at that point

- **Line**

Partial derivative $f(x,y) = x^2 + y^2 + xy$ with respect to y is

- **None options**

Spherical coordinate θ is related to the cylindrical coordinate as _____

- **$\theta = \phi$**

Which of the following is a two dimensional coordinate

- **system Polar coordinate system**

AL-JUNAID INSTITUTE GROUP

Which coordinate system uses two distance and one angle?

- Cylindrical

A plane $z=3$ is _____ to XY-plane

- Parallel

Which of the following function is not the function of one variable

- $M=g(s)+\text{constant}$

XZ-plane consists of all points where

- $x=0$

A zero vector has magnitude equal to zero and _____

- direction No

Which condition is necessary for a function to be continuous at any point (x,y)

- All options

If $w = x + y$ while $x=r + s$ and $y = r-s$ then which of the following is partial derivative of w is w.r.t r ?

- 1

If g and h are continuous functions of one variable, then $f(x,y) = g(x) h(y)$ is a _____ function of x and y .

- Continuous

AL-JUNAID INSTITUTE GROUP

The function of the form $f(x,y) = 3x^2y^5$ is _____ in the domain

- continues everywhere

If the velocity (v_1) of a car is 20m/s eastward and velocity (v_2) of bike is 40 m/s northward then which of the following would be the dot product of these velocities

- -800

Suppose that $z=(x,y)$ is a function of two variable x and y , then partial derivative of function z with respect to x represents the _____ of the tangent of the curve

- None OPTIONS

The product of continuous functions is continuous

- True

The function e^{xy} is continuous In whole region of

- xy plane

The partial derivative of $f(x,y)=x^2y^2$ with respect to x is _____

- x^2e^{-y}

Let $f(x,y) = e^{2y} \cos x$ then $f_x(x,y)=$

- $e^{2y} \sin x$

AL-JUNAID INSTITUTE

GROUP

If $y=f(x)$ and $x=g(t)$ then by chain rule $dy/dt=$

- $Dy/dx * dx/dt$

The partial derivative with respect to y of $f(x,y) = x^2 + 2y^2 + xy$ is _____

- $4y+x$

The mixed 2nd order partial derivatives of $z=xy$ is equal to

xy

Scalar Triple Product of vectors $A=[1,0,0]$, $B=[0,1,0]$ and $C=[0,0,1]$ is-----

- $1(\text{not sure})$

he order of differentiation in an n th order partial derivative can be change without affecting the final result whenever the function and all its partial derivatives of order less than or equal to ' n ' are -----.

- **Differentiable (not sure)**

Let $f(x,y)$ be a function with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0) and,

let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$. If $D=0$, then-----

- $f(x,y)$ has a saddle point at (x_0, y_0) . (not sure)

Geometrically the magnitude of Scalar Triple Product gives the-----.

- **Volume of Parallelepiped**

Value of money is an example of ----- quantity

- **Scalar**

Let $f(x,y)$ be a function with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0) and,

Let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then

- $f(x,y)$ has a saddle point at (x_0, y_0) . (not sure)

AL-JUNAID INSTITUTE

GROUP

Two vectors of magnitudes [4,5] making an angle of 30degree, then the area of parallelogram associated with these two vectors is -----

- Square units

For the function $z=f(x, y, z)$, the total differential dz is equal to

- $dw = df = f_x(x, y, z)dx + f_y(x, y, z)dy + f_z(x, y, z)dz$

For the function $z=f(x, y, z)$, the total differential dz is equal to

- $dz = df = f_x(x, y)dx + f_y(x, y)dy$

vulmsheelp.com